

# Learning Objectives (1/2)

- Parallel Algorithms

- ▶ Familiar with parallel patterns such as reduce, scan, and tiling and can apply them to common parallel programming problems.
- ▶ Can describe parallel algorithms for matrix operations, sorting, dynamic programming, and process coordination.

- Parallel Architectures

- ▶ Can describe shared-memory, message-passing, and SIMD architectures.
- ▶ Can describe a simple cache-coherence protocol.
- ▶ Can identify how communication latency and bandwidth are limited by physical constraints in these architectures.
- ▶ Can describe the difference between bandwidth and inverse latency, and how these impact parallel architectures.

# Learning Objectives (2/2)

- Parallel Performance

- ▶ Understands the concept of “speed-up”: can calculate it from simple execution models or measured execution times.
- ▶ Can identify key bottlenecks for parallel program performance including communication latency and bandwidth, synchronization overhead, and intrinsically sequential code.

- Parallel Programming Frameworks

- ▶ Can implement simple parallel programs in Erlang and CUDA.
- ▶ Can describe the differences between these paradigms.
- ▶ Can identify when one of these paradigms is particularly well-suited (or badly suited) for a particular application.

# Grades

**Heads-up:** This might change (a little).

- Homework 35%
- Midterms: 30%
  - ▶ Two midterms, each worth 15%.
  - ▶ Midterm 1: October 15, in-class
  - ▶ Midterm 2: November 19, in-class
- Final: 35%
- PIKAs (described below): Extra-credit equal to 20% of what you missed on HW and the midterm.
- See Slide 38 for the detailed formulas.

# PIKAs – Pre-Class Activities

- Worth 20% of points missed from HW and the midterm exams
  - ▶ If your pre-final grade is 90%, you can raise that at most 2% through the pikas. Missing one or two isn't a big deal.
  - ▶ If your pre-final grade is 65%, you can raise that 7% from the pikas. This could move your letter grade from a C+ to B).
  - ▶ If your pre-final grade is 45%, you can raise that 11% from the pikas. So do the pikas – I hate turning in failing grades.
  - ▶ See Slide 39 for the detailed formula.
- [PIKA 1](#) is released and is due before lecture on September 5 at 1pm.
- Why **pika**?



- [Pica](#) is a serious eating disorder – if you suffer from an eating disorder, it's an important medical issue. Get help. [Vancouver Coastal Health](#) has excellent programs.
- A [pika](#) is a cute (but unfortunately endangered) mammal that is native to British Columbia.

Photo from [https://www.nwf.org/-/media/NEW-WEBSITE/Shared-Folder/Wildlife/Mammals/mammal\\_american-pika\\_600x300.ashx](https://www.nwf.org/-/media/NEW-WEBSITE/Shared-Folder/Wildlife/Mammals/mammal_american-pika_600x300.ashx).

# Early Bird Bonuses

- Late policy: Late homework will not be accepted.
- Early policy: Early homework gets a 5% bonus.
  - ▶ Typically, the early-bird deadline is two days before the hard deadline.
- No worms.

# Plagiarism

- I have a very simple criterion for plagiarism:  
Submitting the work of another person, whether that be another student, something from a book, or something off the web and representing it as your own is plagiarism and constitutes academic misconduct.
- If the source is clearly cited, then it is not academic misconduct.  
If you tell me “This is copied word for word from Jane Foo’s solution” that is not academic misconduct. It will be graded as one solution for two people and each will get half credit. I guess that you could try telling me how much credit each of you should get, but I’ve never had anyone try this before.
- I encourage you to discuss the homework problems with each other.  
If you’re brainstorming with some friends and the key idea for a solution comes up, that’s OK. In this case, add a note to your solution that lists who you collaborated with.
- More details at:
  - ▶ <http://www.students.cs.ubc.ca/~cs-418/plagiarism.html>
  - ▶ <https://academicintegrity.ubc.ca/regulation-process/students/>

# Collaboration is good

- Talk to your friends.
- Talk with your enemies, if this class brings peace between you, all the better.
- Figure things out together, but write your own code, write your own solutions, **and clearly state who you worked with.**
  - ▶ It's good to collaborate.
  - ▶ List your collaborators at the beginning of each solution you submit.
- I'll give full credit if you work out your own solution after discussing the problem with others.
- If you and a friend work out one solution, turn in two copies of the same solution, **and clearly state that you did so,**
  - ▶ This is not academic misconduct. No prosecution.
  - ▶ Of course, we'll only give credit for one solution; so, you'll each get half.
  - ▶ I've never encountered this scenario in real life.

# Bug Bounties

- If I make a mistake when stating a homework problem, then the **first** person to report the error gets extra credit.
  - ▶ If the error would have prevented solving the problem, then the extra credit is the same as the value of the problem. Post these to piazza or canvas (whichever is chosen in the in-class vote).
  - ▶ Smaller errors get extra credit in proportion to their severity. Send these to me by e-mail. I'm happy to tidy up minor typos, but I've found that the rest of the class doesn't appreciate having posts from the grammar police on the discussion boards.
- The midterm and final have bug bounties awarded in midterm and final exam points respectively.
- The first person to post a bug gets the bounty.



# Mark Can't Hear (very well)

- I will ask you to repeat questions or comments.
- I'll also run around the class so I can hear (and see) you better.
  - ▶ Exercise is healthy.
- Cathy will repeat stuff I miss.

# The Waitlist

- Good news! With a last-minute second-section of CPSC 418 this term, the waitlist is much shorter than usual.
- With usual turnover, I'll guess that most of the waitlist will make it in.
  - ▶ We'll try to squeeze in everyone we can.
- We will **not** exempt waitlist students from HW1 and the first four PIKAs.
  - ▶ **If you want to succeed in this class, do the homework.**
- If you're desperate to get into CPSC 418 to graduate: talk to an advisor.
  - ▶ Course instructors don't make waitlist decisions
  - ▶ The only reason you *might* get wedged in is if this is the only way you can get the last three credits you need to graduate.
  - ▶ It's a department wide optimization problem.